

Maxwell's Equations: Structure, Meaning, and Consequences

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Presentation map

- 1 The Maxwell System
- 2 Physical Meaning of the Four Laws
- 3 Main Consequences
- 4 Takeaways

Source and scope

These slides condense the main ideas from the Maxwell notes: the four laws, their physical meaning, the role of displacement current, charge conservation, potentials, wave propagation, and standard boundary results.

Why Maxwell's Equations Matter

Maxwell's equations are the central laws of classical electromagnetism. They unify

- electrostatics,
- magnetostatics,
- electromagnetic induction,
- charge conservation,
- and the wave nature of light.

In vacuum, the fields are governed by

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

Big idea

The first pair describes sources and constraints; the second pair describes dynamical coupling between electric and magnetic fields.

The Four Equations: Differential and Integral Forms

Differential form

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

Gauss's law for electricity

$$\nabla \cdot \mathbf{B} = 0$$

Gauss's law for magnetism

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Faraday's law

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

Ampère–Maxwell law

Integral form

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0},$$

$$\oint_S \mathbf{B} \cdot d\mathbf{a} = 0,$$

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a},$$

$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \frac{d}{dt} \int_S \mathbf{E} \cdot d\mathbf{a}.$$

Gauss's Laws: Flux, Sources, and No Magnetic Monopoles

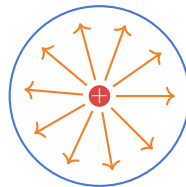
Electric field

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

Electric charge is the source of electric flux. For a point charge, the flux through any enclosing sphere is always q/ϵ_0 .

Locally,

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}.$$



closed surface

Magnetic field

$$\oint_S \mathbf{B} \cdot d\mathbf{a} = 0$$

Magnetic field lines do not begin or end on isolated charges; they form closed loops.

Faraday's Law: A Changing Magnetic Field Creates Circulation in $\mathbf{E}$

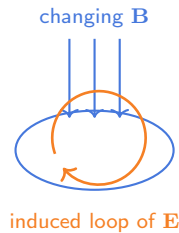
A time-dependent magnetic flux induces an electric field with nonzero circulation:

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}.$$

The minus sign is Lenz's law: the induced response opposes the change in flux.

In differential form,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$



Key consequence

In time-dependent situations, the electric field is generally *not* purely conservative.

Ampère's Law, Maxwell's Correction, and the Capacitor Argument

The magnetostatic law

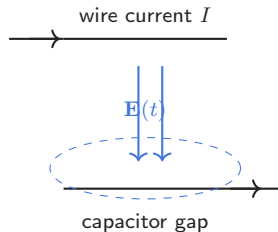
$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

works for steady currents, but it fails in general because charge conservation requires

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.$$

Maxwell fixed the inconsistency by adding the displacement-current term:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$



$$I_D = \epsilon_0 \frac{d\Phi_E}{dt}$$

For an ideal parallel-plate capacitor,
 $I_D = I.$

Interpretation

A changing electric field acts as a source of magnetic

Charge Conservation and the Electromagnetic Potentials

Continuity equation

Taking the divergence of the Ampère–Maxwell law and using Gauss's law gives

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.$$

So charge conservation is built into the full Maxwell system.

Potentials

Because

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

one may write

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t}.$$

The potentials are not unique: gauge transformations leave the physical fields unchanged.

Conceptual point

Maxwell's equations are physical laws; once assumed, they mathematically imply continuity, potentials, boundary conditions, and wave propagation.

Electromagnetic Waves in Vacuum

Set $\rho = 0$ and $\mathbf{J} = 0$. Then the coupled curl equations lead to

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0, \quad \nabla^2 \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} = 0.$$

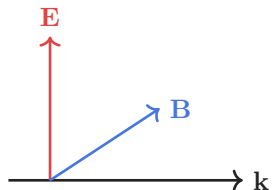
These are wave equations with speed

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}.$$

Maxwell's great prediction was that light itself is an electromagnetic wave.

For a plane wave,

$$\mathbf{B}_0 = \frac{1}{\omega} \mathbf{k} \times \mathbf{E}_0, \quad B_0 = \frac{E_0}{c}.$$



Geometry

The fields are transverse: $\mathbf{E}$, $\mathbf{B}$, and the propagation direction are mutually perpendicular.

Energy Flow, Boundary Conditions, and Matter

Poynting theorem

$$u = \frac{\varepsilon_0}{2} E^2 + \frac{1}{2\mu_0} B^2$$

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}$$

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}$$

Fields carry and transport energy.

Boundary conditions

$$\hat{\mathbf{n}} \cdot (\mathbf{E}_2 - \mathbf{E}_1) = \frac{\sigma}{\varepsilon_0}$$

$$\hat{\mathbf{n}} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0$$

$$\hat{\mathbf{n}} \times (\mathbf{E}_2 - \mathbf{E}_1) = \mathbf{0}$$

$$\hat{\mathbf{n}} \times (\mathbf{B}_2 - \mathbf{B}_1) = \mu_0 \mathbf{K}$$

Macroscopic matter form

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}, \quad \mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}$$

$$\nabla \cdot \mathbf{D} = \rho_f, \quad \nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}$$

These forms separate free sources from material response.

Takeaways and Final Synthesis

- Gauss's laws encode electric sources and the absence of magnetic monopoles.
- Faraday's law and the Ampère–Maxwell law describe the mutual induction of $\mathbf{E}$ and $\mathbf{B}$.
- The displacement current is essential for consistency with charge conservation.
- The full system implies potentials, electromagnetic waves, energy flow, and interface conditions.
- In relativity, the electric and magnetic fields become components of a single field tensor.

Final takeaway

What began historically as separate electric and magnetic laws becomes, through Maxwell, one unified dynamical field theory.

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu, \quad \partial_\nu J^\nu = 0.$$